

Molecular Modelling Course

November 18. 2025 – 20. 2025 | Dr. Birkan Emrem, Dr. Ferdinand Jamitzky, Dr. Plamen Dobrev, Dr. Prasanth Ganta, Dr. Elmira Birang
Leibniz Supercomputing Centre

Outline



- Motivation
- Bio-molecular atomistic structures
- Molecular dynamics introduction
- Introduction to GROMACS
- Input file parameters and topologies for molecular dynamics simulations and in GROMACS in particular

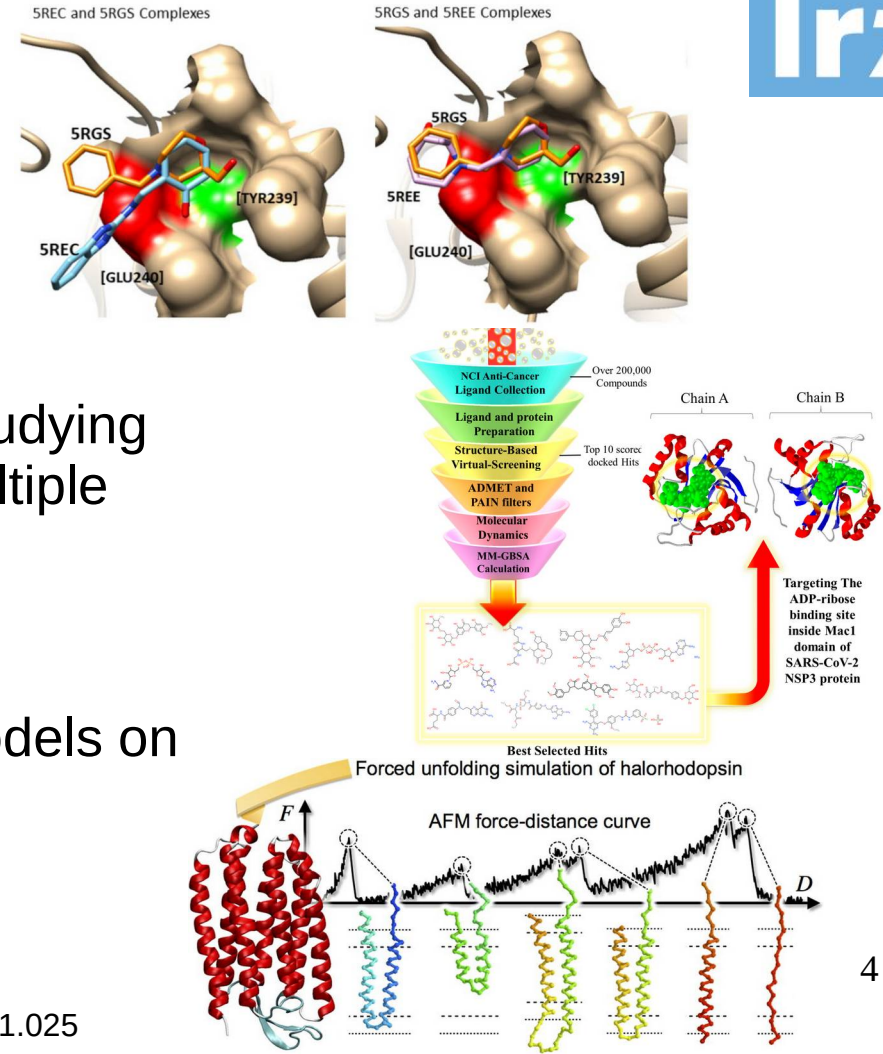
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Why do simulations?

- To study observables not accessible by experimental approaches
- To carry out high throughput essays, studying system under multiple conditions or multiple systems simultaneously
- Reproduce experiment and validate models on microscopic level

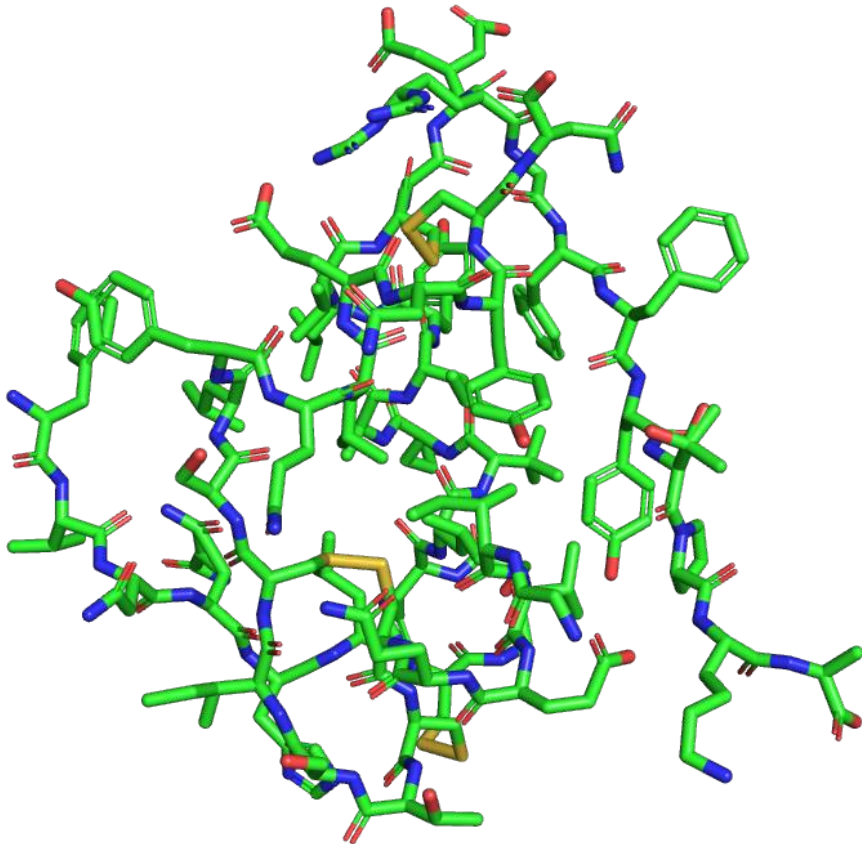


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- **Bio-molecular atomistic structures**
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Biomolecular Structures



- Structures of biological molecules are saved as a set of atomic coordinates
- They are derived experimentally either by X-ray crystallography, atomic NMR or Cryo-EM, or *in silico* using AI methods (e.g. AlphaFold)
- Atomic resolution of a structure is crucial for obtaining physiologically relevant data during simulation

Redesigned PDB Statistics Support Enhanced Functionality

Explore Statistics

Welcome

Deposit

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Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

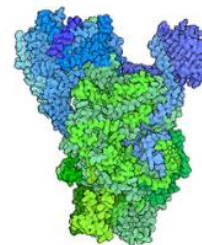
Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive

Computed Structure Models (CSM) from AlphaFold DB and ModelArchive

These data can be explored in context of external annotations providing a structural view of biology.



May Molecule of the Month



TOC-TIC Translocon

Latest Entries

As of Tue May 13 2025



9EHY

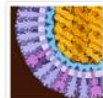
X-ray crystal structure of ADC-33 beta-lactamase in complex with cefazidime in acyl and product forms

Features & Highlights



Announcement: Ligand Expo To Be Retired in 2025

Users should transition to RCSB PDB and wwPDB services as soon as possible



Watch the Webinar: Seeing Bird Flu in 3D

This virtual course will help you explore Influenza A H5N1 virus protein structures using RCSB.org tools



Watch the Webinar: Streamlining Access to RCSB PDB APIs with Python

This virtual course introduces and demonstrates usage of the rcsb-api Python package for easy interactions with RCSB PDB Search and Data API services

News

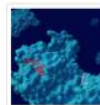
Publications



Access Computed Structure Model Annotations

Gene Ontology terms (for Molecular Function, Cellular Component, and Biological Process), InterPro protein family classifications, and Pharos disease associations are available for CSMs at RCSB.org

05/19/2025



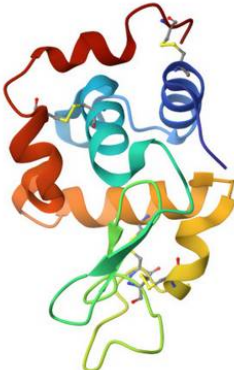
PDB-101 Focus: Peak Performance

PDB-101 materials explore the structural biology of nutrition.

05/16/2025



Paper Published: A new chapter for Molecule of the Month



[Explore in 3D](#): [Structure](#) | [Sequence Annotations](#) | [Electron Density](#) | [Validation Report](#)

Global Symmetry: Asymmetric - C1 
Global Stoichiometry: Monomer - A1 

[Find Similar Assemblies](#)

Biological assembly 1 assigned by authors.

Macromolecule Content 

- Total Structure Weight: 14.33 kDa 
- Atom Count: 1,082 
- Modeled Residue Count: 129 
- Deposited Residue Count: 129 

2YVB | [pdb_00002yvb](#)

High resolution X-ray crystal structure of Tetragonal Hen egg white lysozyme

PDB DOI: <https://doi.org/10.2210/pdb2YVB/pdb>

Classification: [HYDROLASE](#)

Organism(s): [Gallus gallus](#)

Mutation(s): No 

Deposited: 2007-04-10 **Released:** 2007-04-24

Deposition Author(s): [Naresh, M.D.](#), [Jaimohan, S.M.](#), [Kumar, V.V.](#), [Mandal, A.B.](#)

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 1.62 Å

R-Value Free:
0.277 (Depositor), 0.280 (DCC) 

R-Value Work:
0.229 (Depositor), 0.230 (DCC) 

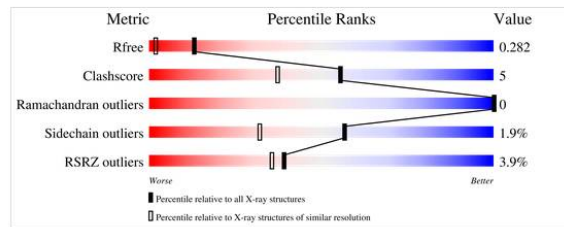
R-Value Observed:
0.231 (Depositor) 

Starting Model: experimental

[View more details](#)

wwPDB Validation

[3D Report](#) | [Full Report](#)



This is version 1.6 of the entry. See complete [history](#).

Literature

[Download Primary Citation](#) 

High resolution X-ray crystal structure of Tetragonal Hen egg white lysozyme

[Naresh, M.D.](#), [Jaimohan, S.M.](#), [Kumar, V.V.](#)

To be published.

PDB file format



		GRowing Old MAKes el Chrono Sweat										
		THIS IS A SIMULATION BOX										
		CRYST1	50.000	50.000	50.000	90.00	90.00	90.00	P 1	1		
		MODEL	1									
atom number		ATOM	1	N	GLY A	1	26.489	31.416	34.421	1.00	21.88	N
		ATOM	2	CA	GLY A	1	25.423	31.498	33.376	1.00	22.85	C
		ATOM	3	C	GLY A	1	25.301	30.097	32.750	1.00	43.92	C
atom name		ATOM	4	O	GLY A	1	25.570	29.200	33.539	1.00	25.22	O
		ATOM	5	N	ILE A	2	25.019	30.003	31.464	1.00	26.28	N
		ATOM	6	CA	ILE A	2	24.864	28.738	30.732	1.00	20.84	C
residue name		ATOM	7	C	ILE A	2	25.985	27.774	30.790	1.00	11.81	C
		ATOM	8	O	ILE A	2	25.772	26.564	31.101	1.00	20.31	O
		ATOM	9	CB	ILE A	2	24.469	28.965	29.227	1.00	40.00	C
chain identifier		ATOM	10	CG1	ILE A	2	23.773	27.618	28.829	1.00	36.74	C
		ATOM	11	CG2	ILE A	2	25.611	29.266	28.272	1.00	23.02	C
		ATOM	12	CD1	ILE A	2	22.539	27.503	29.772	1.00	26.69	C
residue number		ATOM	13	N	VAL A	3	27.219	28.231	30.615	1.00	16.57	N
		ATOM	14	CA	VAL A	3	28.386	27.373	30.708	1.00	15.75	C
		ATOM	15	C	VAL A	3	28.460	26.633	32.054	1.00	22.09	C
coordinates	X	ATOM	16	O	VAL A	3	28.805	25.462	32.169	1.00	24.52	O
		ATOM	17	CB	VAL A	3	29.655	28.180	30.357	1.00	21.34	C
		ATOM	18	CG1	VAL A	3	30.970	27.432	30.580	1.00	32.48	C
	Y	ATOM	19	CG2	VAL A	3	29.510	28.681	28.909	1.00	26.35	C
		ATOM	20	N	GLU A	4	28.309	27.491	33.067	1.00	16.58	N
		ATOM	21	CA	GLU A	4	28.463	26.946	34.427	1.00	15.32	C
		ATOM	22	C	GLU A	4	27.348	26.030	34.742	1.00	16.88	C
	Z	ATOM	23	O	GLU A	4	27.464	24.946	35.260	1.00	23.30	O
		ATOM	24	CB	GLU A	4	28.543	28.163	35.398	1.00	17.11	C
occupancy		ATOM	25	CG	GLU A	4	29.737	29.037	35.083	1.00	21.45	C
		ATOM	26	CD	GLU A	4	29.648	29.929	33.867	1.00	21.59	C
		ATOM	27	OE1	GLU A	4	28.595	30.431	33.509	1.00	23.43	O
B-factor		ATOM	28	OE2	GLU A	4	30.784	30.041	33.311	1.00	25.36	O

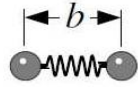
Outline



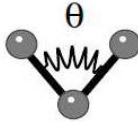
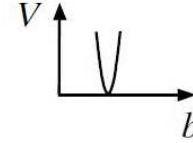
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Molecular Dynamics Introduction

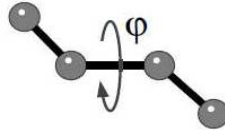
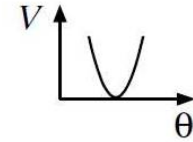
classic potentials



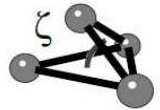
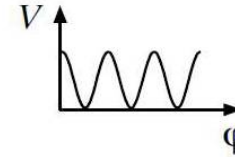
$$V^B = \frac{1}{2}k_b(b - b_0)^2$$



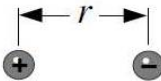
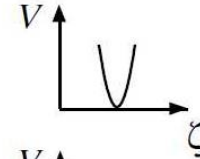
$$V^W = \frac{1}{2}k_\theta(\theta - \theta_0)^2$$



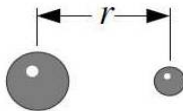
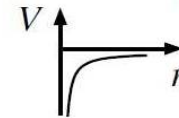
$$V^D = k_\varphi[1 + \cos(n\varphi - \delta)]$$



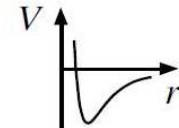
$$V^E = \frac{1}{2}k_\zeta(\zeta - \zeta_0)^2$$



$$V^C = q_i q_j / (4\pi\epsilon_0\epsilon_r r)$$

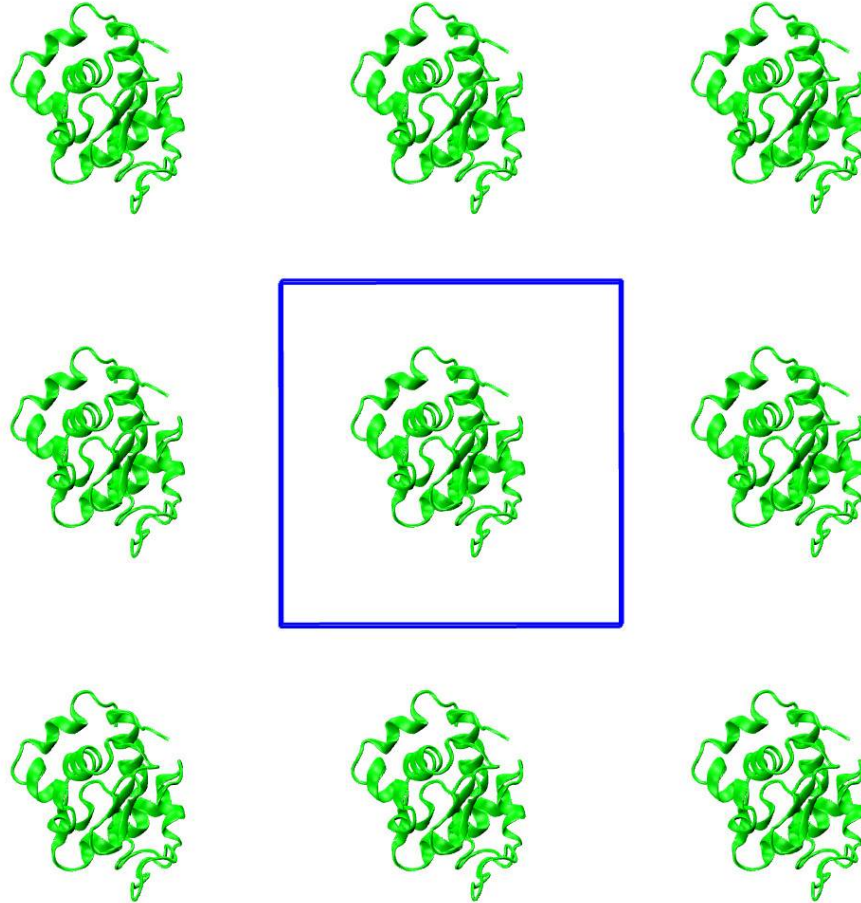


$$V^{LJ} = C_{12}(i, j)/r^{12} - C_6(i, j)/r^6$$



Molecular Dynamics Introduction

periodic boundaries



Molecular Dynamics Introduction

thermostats and barostats



Berendsen thermostat

$$\frac{dT}{dt} = \frac{T_0 - T}{\tau}$$

$$\lambda = \left[1 + \frac{n_{TC} \Delta t}{\tau_T} \left\{ \frac{T_0}{T\left(t - \frac{1}{2} \Delta t\right)} - 1 \right\} \right]^{1/2}$$

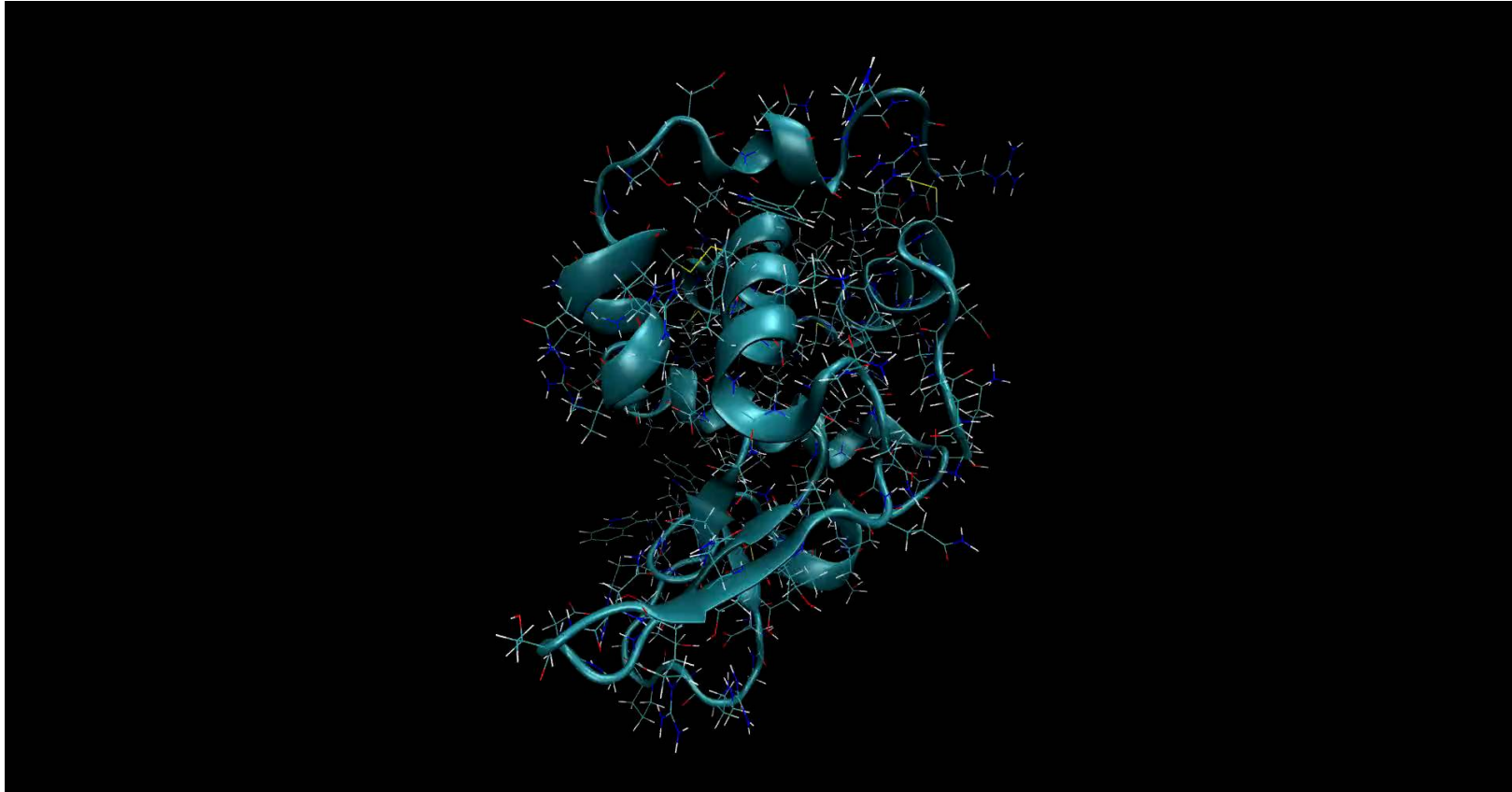
Berendsen barostat

$$\frac{dP}{dt} = \frac{P_0 - P}{\tau_p}$$

$$\mu_{ij} = \delta_{ij} - \frac{n_{PC} \Delta t}{3 \tau_p} \beta_{ij} \{ P_{0ij} - P_{ij}(t) \}$$

Molecular Dynamics Introduction

trajectories



Outline

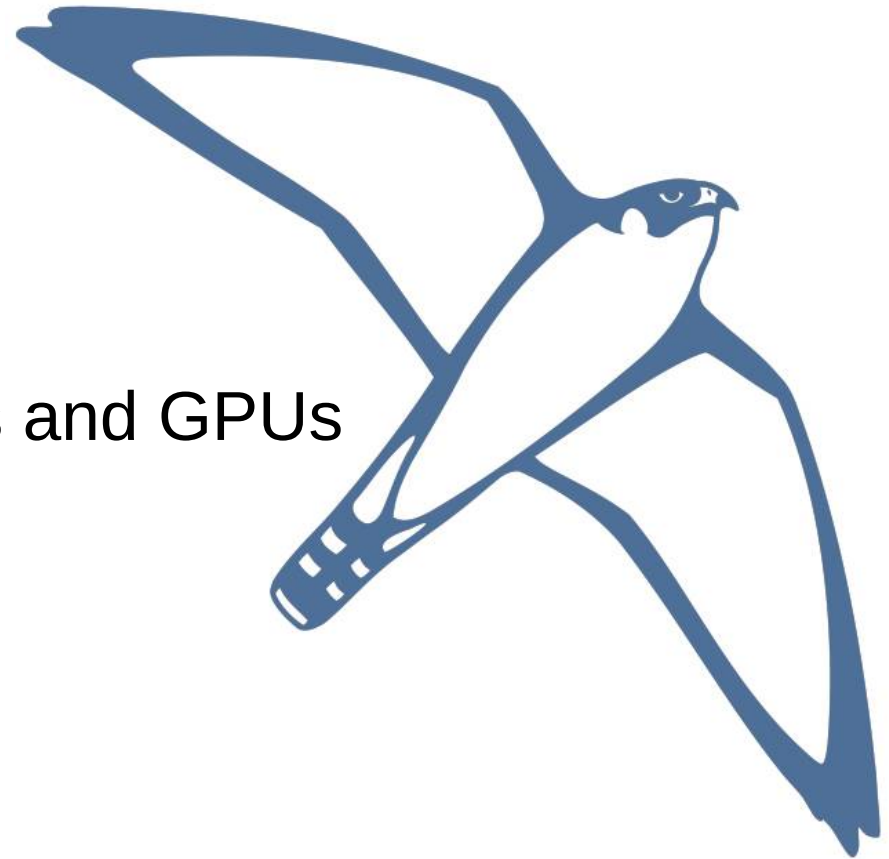


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- **Introduction to GROMACS**
- Input file parameters and topologies for molecular dynamics simulations and in GROMACS in particular

Introduction to GROMACS



- Open Source
- Highly parallel
- Supports almost all modern CPUs and GPUs
- Easy and straightforward to use



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Topologies



Molecular descriptions in MD software packages are stored in topologies

Introduction to GROMACS

topologies



```
; Include forcefield parameters
#include "amber99sb-ildn.ff/forcefield.itp"
```

```
[ moleculetype ]
; Name          nrexcl
Protein_chain_A 3
```

```
[ atoms ]
; nr  type  resnr residue  atom  cgnr  charge  mass  typeB  chargeB  massB
; residue 1 LYS rtp NLYS q +2.0
 1    N3    1    LYS    N     1    0.0966  14.01
 2    H     1    LYS    H1    2    0.2165   1.008
 3    H     1    LYS    H2    3    0.2165   1.008
 4    H     1    LYS    H3    4    0.2165   1.008
 5    CT    1    LYS    CA    5   -0.0015  12.01
 6    HP    1    LYS    HA    6    0.118    1.008
 7    CT    1    LYS    CB    7    0.0212  12.01
 8    HC    1    LYS    HB1   8    0.0283   1.008
 9    HC    1    LYS    HB2   9    0.0283   1.008
10    CT    1    LYS    CG   10   -0.0048  12.01
11    HC    1    LYS    HG1  11    0.0121   1.008
12    HC    1    LYS    HG2  12    0.0121   1.008
13    CT    1    LYS    CD   13   -0.0608  12.01
14    HC    1    LYS    HD1  14    0.0633   1.008
15    HC    1    LYS    HD2  15    0.0633   1.008
16    CT    1    LYS    CE   16   -0.0181  12.01
17    HP    1    LYS    HE1  17    0.1171   1.008
```

```
[ bonds ]
; ai  aj  funct      c0      c1      c2      c3
 1    2    1
 1    3    1
 1    4    1
 1    5    1
 5    6    1
 5    7    1
 5   23    1
 7    8    1
 7    9    1
 7   10    1
10   11    1
10   12    1
10   13    1
```

Introduction to GROMACS

topologies



```
[ angles ]
; ai  aj  ak funct      c0      c1      c2      c3
  2   1   3   1
  2   1   4   1
  2   1   5   1
  3   1   4   1
  3   1   5   1
  4   1   5   1
  1   5   6   1
  1   5   7   1
  1   5  23   1
  6   5   7   1
```

```
[ dihedrals ]
; ai  aj  ak  al funct      c0      c1      c2      c3      c4      c5
  2   1   5   6   9
  2   1   5   7   9
  2   1   5  23   9
  3   1   5   6   9
  3   1   5   7   9
  3   1   5  23   9
  4   1   5   6   9
```

Introduction to GROMACS

topologies



```
; Include Position restraint file
#ifdef POSRES
#include "posre.itp"
#endif

; Include water topology
#include "amber99sb-ildn.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water oxygen
[ position_restraints ]
; i funct      fcx      fcy      fcz
  1  1      1000      1000      1000
#endif

; Include topology for ions
#include "amber99sb-ildn.ff/ions.itp"

[ system ]
; Name
LYSOZYME C

[ molecules ]
; Compound      #mols
Protein_chain_A  1
SOL              6628
NA               20
CL               28
```


Introduction to GROMACS

topologies



```
; Include forcefield parameters
#include
"amber99sb-ildn.ff/forcefield.itp"

; Include chain topologies
#include "topol_Protein_chain_A.itp"
#include "topol_Protein_chain_B.itp"

; Include water topology
#include "amber99sb-ildn.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water
oxygen
[ position_restraints ]
; i funct      fcx      fcy      fcz
  1  1      1000      1000      1000
#endif

; Include topology for ions
#include "amber99sb-ildn.ff/ions.itp"

[ system ]
; Name
INSULIN (CHAIN A); INSULIN (CHAIN
B)

[ molecules ]
; Compound      #mols
Protein_chain_A    1
Protein_chain_B    1
SOL                 31
```

Force field organization



The potentials among all atom types inside a protein are described by set of parameters called force field

Introduction to GROMACS

force fields

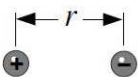


Molecule definition

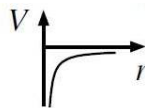
```
[ALA]
[ atoms ]
  N  N      -0.41570  1
  H  H      0.27190  2
  CA CT      0.03370  3
  HA H1      0.08230  4
  CB CT     -0.18250  5
  HB1 HC      0.06030  6
  HB2 HC      0.06030  7
  HB3 HC      0.06030  8
  C  C       0.59730  9
  O  O      -0.56790 10

[ bonds ]
  N  H
  N  CA
  CA HA
  CA CB
  CA  C
  CB HB1
  CB HB2
  CB HB3
  C  O
 -C  N

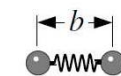
[ impropers ]
 -C  CA  N  H
  CA +N  C  O
```



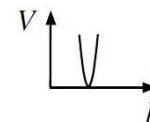
$$V^C = q_i q_j / (4\pi\epsilon_0\epsilon_r r)$$



Bonded parameters

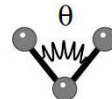


$$V^B = \frac{1}{2}k_b(b - b_0)^2$$

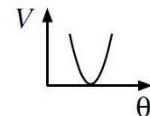


[bondtypes]

```
; i j func b0 kb
C C 1 0.1525 259408.0 ; new99
C OS 1 0.1323 376560.0 ; new99
C H4 1 0.1080 307105.6 ; new99
```

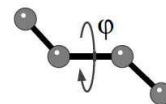


$$V^W = \frac{1}{2}k_\theta(\theta - \theta_0)^2$$

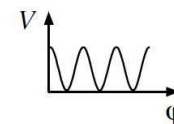


[angletypes]

```
; i j k func th0 cth
HW OW HW 1 104.520 836.800 ; TIP3P water
HW HW OW 1 127.740 0.000 ;
C C O 1 120.000 669.440 ; new99
C C OH 1 120.000 669.440 ; new99
CT C CT 1 117.000 527.184 ; new99
```



$$V^D = k_\varphi[1 + \cos(n\varphi - \delta)]$$

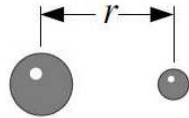


[dihedraltypes]

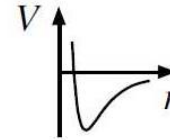
```
; i j k l func
CT CT OS CT 9 0.0 1.60247 3 ;
CT CT OS CT 9 180.0 0.41840 2 ;
C N CT C 9 0.0 1.12968 2 ; new for 99sb
C N CT C 9 0.0 1.75728 3 ; new for 99sb
```

Introduction to GROMACS

force fields



$$V^{\text{LJ}} = C_{12}(i,j)/r^{12} - C_6(i,j)/r^6$$



[atomtypes]

```
; name      at.num  mass    charge ptype sigma    epsilon  
Br          35     79.90   0.0000  A   3.95559e-01 1.33888e+00 ; Converted from parm99.dat  
C           6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CA          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CB          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CC          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CK          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CM          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CN          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CQ          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CR          6      12.01   0.0000  A   3.39967e-01 3.59824e-01  
CT          6      12.01   0.0000  A   3.39967e-01 4.57730e-01  
CV          6      12.01   0.0000  A   3.39967e-01 3.59824e-01
```

Input parameters



Specifying the way the simulation would run
and output data is described by the input
parameters

Introduction to GROMACS

simulation parameters



```
define          = -DPOSRES
;disre          = simple
;disre_fc       = 20
constraints     = h-bonds
constraint-algorithm = Lincs
lincs-order     = 6
integrator      = md
dt              = 0.004 ; ps !
comm_mode       = Linear
nsteps          = 2500000000;000000 ; total 10000 ps.
nstcomm         = 1

nstxout         = 100000
nstvout         = 100000
nstfout        = 0

nstxtcout = 5000

nstlog          = 1000
nstenergy       = 1000
nstlist         = 5
ns_type         = grid
rlist           = 1
vdwtype         = Cut-off
cutoff-scheme   = Verlet
rvdw            = 1
coulombtype     = PME
fourierspacing  = 0.12
pme_order       = 4
rcoulomb        = 1
```

Introduction to GROMACS

simulation parameters



; Berendsen temperature coupling is on in
two groups

```
Tcoupl      = v-rescale  
tau_t       = 0.1      ; 0.5
```

```
tc-grps     = System   ;  
Protein SOL_NA+_CL-  
ref_t       = 300      ; 300
```

; Pressure coupling is not on
Pcoupl = c-rescale

```
pcoupltype = Isotropic  
tau_p      = 1  
compressibility = 4.5e-5  
ref_p      = 1.0
```

; Generate velocities is on at 300 K.

```
gen_vel     = yes  
gen_temp    = 300.0  
gen_seed    = -1
```

Introduction to GROMACS

simulation parameters, energy minimization



```
define          = -DFLEXIBLE
constraints      = none
integrator       = steep
nsteps          = 100
;
;               Energy minimizing stuff
;
emtol           = 2000
emstep          = 0.01

nstcomm         = 1
ns_type         = grid
rlist           = 1
rcoulomb        = 1.0
rvdw            = 1.0
Tcoupl          = no
Pcoupl          = no
gen_vel         = no
```